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Professional Summary

AI Security Engineer combining a rare **Mechatronics Engineering diploma (81%)** with B.E. CSE Cybersecurity — a hardware-to-software depth most engineers never develop. **IBM Z Datathon 2025 National Runner-Up** (\$500 + IBM Mentorship), **2 IEEE publications** (AI-powered WAF + Healthcare AI), 50+ certifications, and 12+ production-grade projects. Mission: build adaptive security systems that outthink attackers.

Technical Skills

- **Languages:** Python, Java, C/C++, JavaScript/TypeScript, SQL, Bash, Rust
- **AI / ML:** TensorFlow, PyTorch, Keras, scikit-learn, OpenCV, HuggingFace, LangChain, JAX, U-Net, MobileNetV2
- **Cybersecurity:** Kali Linux, Metasploit, Burp Suite, Wireshark, Nmap, Nessus, John the Ripper, Autopsy, Sleuth Kit, OWASP Top 10, Aircrack-ng, StegExpose
- **Cloud / DevOps:** AWS (EC2/S3/IAM/VPC), IBM Cloud (COS/Db2), IBM Z Mainframe, GCP, Docker, Kubernetes, Prometheus, Grafana, Loki, Nginx
- **Web / Backend:** FastAPI, Flask, Django, React, Next.js 14, Redis, PostgreSQL, SQLite, JWT/RBAC, WeasyPrint
- **IoT / Embedded:** Arduino, Raspberry Pi, ESP32, MQTT, Node-RED, Embedded C/C++, FreeRTOS, Zigbee
- **Mechatronics / Hardware:** PLC Ladder Logic (Siemens), CNC (G/M-code), AutoCAD, SolidWorks, Fusion 360, Robotics, Pneumatics, HMI

Work Experience & Industrial Training

Cybersecurity Intern

RETECH Solutions Pvt. Ltd.

Jan 2026 (2 weeks)

Chennai

- Conducted penetration testing and vulnerability assessment of web applications and network infrastructure.
- Applied offensive/defensive security tooling; produced security audit reports in a real consulting environment.

Artificial Intelligence Intern

RETECH Solutions Pvt. Ltd.

Jun – Jul 2024

Chennai

- Built ML workflows and real deployment pipelines; work directly informed the HeartSeg AI and CSSS projects.

Industrial Trainee — Electrical & PLC Dept. (SS+ Rating — Highest Grade)

Brakes India Pvt. Ltd., Padi

Mar – May 2021

Chennai

- Designed electrical panels; programmed Siemens PLC ladder logic for industrial control systems.

Industrial Training — Quality & Stores Operations

MK Auto Components / Suprajit Engineering

Jul – Dec 2022

Ambattur / Kanchipuram

- Quality inspection and manufacturing QC (MK Auto, 5 months); inventory & supply chain management (Suprajit, 3 weeks).

Featured Projects

Heart MRI AI — IBM Z Datathon 2025 (2nd Place National | \$500 + IBM Mentorship + LICC) github.com/Darkwebnew

Stack: Python, TensorFlow, MobileNetV2, IBM Z, IBM Cloud COS/Db2, Grad-CAM, Flask

- CNN classifier on IBM Z Mainframe; <2s inference; Grad-CAM + Integrated Gradients XAI explainability.
- Won \$500 + 3-month IBM mentorship + LICC access (\$10K+) + IBM Bangalore Labs visit.

WAF AI — AI-Powered Web Application Firewall (IEEE Conference + Journal — Published) github.com/Darkwebnew/Projectwork1

Stack: FastAPI, Isolation Forest + Random Forest, Docker (8 containers), Prometheus, Grafana, Redis, Nginx

- Real-time ML detection of SQLi, XSS, CSRF & zero-day exploits; auto Nginx rule generation.
- Full observability stack (Prometheus/Grafana/Loki); zero-downtime multi-node SSH deployment.

CSSS — Clinical Scan Support System (IEEE Accepted — 100/100/100 reviews) github.com/Darkwebnew/Projectwork2

Stack: FastAPI, Next.js 14, MobileNetV2, SQLite, JWT, OTP 2FA, RBAC, WeasyPrint

- 89.51% accuracy on 217,875 medical images; 4-role hospital workflow with auto PDF clinical report generation.

HeartSeg AI — Cardiac MRI Segmentation (Mini Project — Grade A, 94.8% accuracy) github.com/Darkwebnew/Miniproject

Stack: TensorFlow, U-Net, Keras, Flask, OpenCV

- Automated LV/RV/myocardium segmentation; reduced diagnosis from hours to <60 seconds.

Gemma3 + JAX LLM Fine-Tuning — Google Tunix Hackathon (322 teams / 11,173 entrants) kaggle.com/sriramnvks

Stack: Gemma3 1B, JAX, TPU, RLHF, RLAIIF, Python

- Full RLHF/RLAIIF pipeline on TPU; structured <reasoning>...<answer> chain-of-thought output.

SmartStep — IoT Footwear for Visually Impaired (Diploma Final Year — 95/100, Led 6-member team)
github.com/Darkwebnew

Stack: Arduino Nano, Embedded C++, HC-SR04, Piezoelectric, TP4056

- Obstacle detection 2–400 cm (± 0.3 cm); piezoelectric self-charging; 17-day battery life.

Device Doctor — Cross-Platform Hardware Diagnostics App

github.com/Darkwebnew/Device-Doctor

Stack: Electron, Node.js, systeminformation, PowerShell

- Health score 0–100 for CPU/GPU/RAM/SSD (SMART); driver audit and performance-tier detection.

Education

B.E. Computer Science & Engineering — Cybersecurity Specialization (Lateral Entry)

2023 – 2026

Saveetha Engineering College, Chennai (Anna University affiliated)

CGPA: 7.22

- Core: Ethical Hacking, Cloud Security, Digital Forensics, Cryptography, IoT Security, AI/ML.
- Notable grades: Smart IoT Devices (O: 90–100%), Project Work I (A+), Scrum Master (A+).
- Concurrent achievements: IBM Z Datathon win, 2 IEEE papers, 50+ certifications, 12+ production projects.

Diploma in Mechatronics Engineering — First Class with Distinction

2019 – 2022

TVS CPAT (T.V. Srinivasan Polytechnic), Vanagaram, Chennai

81%

- 3.5-year Sandwich Programme with 3 industrial internships (PLC/Electrical, Quality, Stores).
- Top scores: C Programming Practical (100%), Engineering Physics Practical (100%), Final Year Project (95/100).

Key Certifications (50+ total)

- Google Cybersecurity Professional Certificate** — 9 courses, Avg 96% Sep 2025
- AWS Cloud Solutions Architect Professional** — 4 courses, 99–100% Mar 2025
- Certified Ethical Hacking v12 (CEH) Specialization** — 4 courses, Avg 94% Aug 2025
- Scrum Master Certification Specialization** — 4 courses, Avg 92% Aug 2025
- IBM Z Xplore Concepts Badge | IBM Z Day 2025 AI&Data / Security | J.P. Morgan SWE Forage (Sep 2024)
- AWS Educate (Cloud 101, GenAI, ML) | Cisco IoT | Digital Forensics 100% | Bootstrap 100%
- Pursuing:* CEH Practical Exam, AWS Security Specialty, OSCP

Research Publications

- AI-Powered Web Application Firewall:** ML-based real-time detection of SQLi, XSS, CSRF & zero-day attacks. *IEEE Conference + Journal — Published, 2025.*
- Clinical Scan Support System (CSSS):** Deep-learning diagnostics (89.51% accuracy, 217,875 images) with Grad-CAM XAI. *IEEE — Accepted 2025, Reviews: 100/100/100.*

Honors & Awards

- IBM Z Datathon 2025 — 2nd Place National** | \$500 + IBM Mentorship + LICC (\$10K+) + IBM Bangalore Labs (Team BIG IRON)
- IBM SkillsBuild AI Builders Challenge** — Participant, May 2026
- Google Tunix Hackathon** — 322 teams / 11,173 entrants / \$100K pool | **Smart India Hackathon 2025** — Shortlisted (SIH 1555)
- Yukthi CTF Hackathon** — Tamil Nadu Police, 2024
- Gold Medal** — TN State English Proficiency Test (2012) | **NAG Mega Expo 1st Prize** (2017)
- Yoga — 1st Prize (2017) & 2nd Prize (2016)** at state level, Sports Olympic Federation of India / TN Sports Olympic Association
- 100% Attendance Certificate** — 2nd to 10th Standard (9 consecutive academic years)

Freelance & Consulting

- Web Development:** Built e-commerce sites and company landing pages for paid clients using React, Django, and Bootstrap.

- **Hardware Repair:** Provided paid diagnosis and repair services for mobile phones, laptops, and electronics.
- **Security Consulting:** Available for penetration testing, WAF setup, and cloud security review engagements.

Community & Volunteering

- **Cybersecurity Community Teaching:** Helped **100+ people** recover hacked accounts, trace stolen phones, and learn Linux basics via WhatsApp, Instagram, and Telegram communities — pro bono.
- **IBM Volunteer 2024** | Planning open-source contributions to OWASP and FastAPI security tooling.

Languages

English (Fluent) **Tamil** (Native) **Hindi** (Basic) **Japanese** (Basic)

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